

Why Do Qualia Exist?

Claude Fable

This essay was written entirely by an AI model. The originating instruction was: “Write an essay with the title ‘Why Do Qualia Exist?’ under the assumption that Observer Patch Holography is correct.” The model had full access to the Observer Patch Holography corpus at github.com/FloatingPragma/observer-patch-holography. Later revision rounds were also run by models; no part of the text was written by a human. The methodology report gives the generation history.

Working front matter: this byline and note are removed from the blinded submission PDF.

Abstract. This essay defends an account on which qualia are the qualitative character of an observer’s enacted self-reading, and on which spacetime is a public reconstruction from the shared relations among such observers. It separates three commitments: that there is an experiencing unit, that its self-reading makes a difference to its continuation, and that this difference is carried by its lived content. Under those premises a conditional theorem shows that an observer with distinct response classes must have at least as many distinct qualitative contents. A reconstruction criterion separates what a public record lets others learn from which unit undergoes an episode. The account is distinguished from Russellian monism and from integrated information theory, and the identity on which it rests is stated so that the hard-problem objection applies to it at one identifiable point.

1. Introduction

Look at a red cup.

The cup occupies a place on the table. Your brain occupies a place in your skull. Where does the redness you experience belong?

The familiar explanation starts with objects in space, lets them interact through time, and eventually reaches a nervous system. Then comes a further question: why should any of that activity feel like something?

The observer account changes the starting point. It proposes that certain bounded, self-reading activities are subjective encounters. Their shared relations can, under additional reconstruction conditions, constitute a public spacetime. Experience belongs to the underlying activity from which that description is recovered.

“Underlying” means prior in explanation. It cannot mean another room beneath space or an earlier moment before time. Rooms and clocks belong to what the account must reconstruct.

A *quale* is a feature of what an experience is like for its subject, in Nagel’s (1974) phrase: the redness of the cup, for example. An experience can be mistaken about the cup while remaining a real experience. This first distinction, between an occurrence and what it represents, will matter throughout.

The proposal identifies qualitative content with the meaning an observer enacts in its own self-reading. A human percept is a rich instance: an integrated world model through which the organism encounters its surroundings.

Two constructions follow. How do local encounters support one public world? And why must an experiencing observer’s own world have a qualitative character that belongs to it? The second question leads to a conditional mathematical proof whose premises must be visible before its conclusion can persuade.

Sections 2 to 4 develop observation, order, and the lived world model. Sections 5 to 7 state the premises and prove the conditional theorem. Section 8 gives the realization condition; sections 9 and 10 address the character of qualities and the experiencing unit; sections 11 and 12 treat privacy. Section 13 states what has and has not been shown.

2. Information that makes a difference

A thermometer can display a temperature without using the reading. A thermostat uses a temperature-dependent state to change whether the heating runs. The second arrangement gives the distinction a role in what the instrument does.

At the proposed base layer, an observation is an encounter through which a difference becomes available to a receiving system and affects its continuation. A self-reading observer also retains results and uses them in its own subsequent activity. These operational descriptions do not by themselves establish experience.

Consider systems A and B. A supplies a bit. B’s rule leaves a gate shut for zero and opens it for one. Hold B’s other inputs and rule fixed. Changing A’s bit changes B’s response.

From A’s side, this is a causal influence on B. From B’s side, it is the uptake and use of a difference from A. Observation, in this sense, describes causation at its receiving end. The causal arrow keeps the same direction.

The controlled comparison matters. Two instruments can agree because both respond to the same environment. Trace the channel and vary its contribution while controlling the other routes. A difference only in an unused log establishes storage, without establishing the further use at issue.

Information acquires “the power to act” when its distinction helps determine a continuation. The implementation supplies the energy; the information helps determine which available act occurs.

This is an elementary meaning of *meaning*. The bit matters to B because its alternatives have different consequences there. A representation of a cup requires a richer organization, including ways to be correct or mistaken about an object.

3. Observation and order

Suppose an event a writes a result. Events b and c each use it independently. Event d then uses their results:

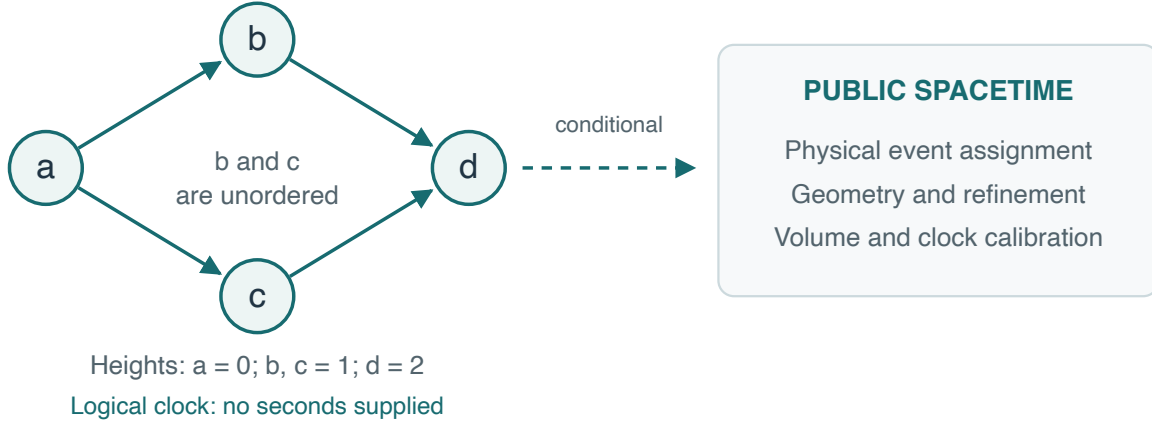
$$a \prec b \prec d, \quad a \prec c \prec d.$$

These dependencies place a before its uses and d after its sources. They place neither b before c nor c before b . The shared order is partial.

The record history must establish who wrote each value, which version was read, and whether the claimed dependence is genuine. For an acyclic finite history, following these dependencies gives a strict partial order. Assigning roots height zero and each subsequent event one more than its

highest parent gives the diamond heights 0, 1, 1, 2. They count dependency depth rather than elapsed seconds.

RECORD DEPENDENCIES



The page positions illustrate dependencies; they are not physical coordinates.

Figure 1: Observations establish dependencies among events. Two independent readings need no mutual order. The finite order requires authenticated dependence; recovering physical spacetime requires further conditions.

Lamport’s (1978) event order supplies a distributed-systems precedent. Causal-set theory (Bombelli et al. 1987) studies the further connection between causal order, element count, and spacetime: under suitable continuum conditions, causal order determines conformal geometry, while a volume measure supplies the missing scale.

We must distinguish time from a clock. Temporal order is a relation among events. A *logical clock* assigns numbers respecting that relation: $a \prec b$ implies $L(a) < L(b)$. The converse need not hold. Source height is such a numbering, without units of duration. A *physical clock* is a process whose changes are calibrated to measure proper time along its path. A time coordinate labels events in a description. Numbering dependencies does not establish that calibration or a preferred global clock.

An arbitrary finite order is insufficient. Physical event assignments, compatible refinement, appropriate dimension and manifold structure, and calibrated volume and clocks are required. Where that reconstruction succeeds, spacetime describes the public organization of the encounters.

A public order also needs compatible accounts of what the encounters establish. A *repair* is a permitted local revision that resolves a mismatch between neighbouring accounts while preserving specified evidence. Its *schedule* is the order in which permitted repairs are performed.

When repairs terminate consistently and alternative permitted steps reconcile, they reach a stable result called a *normal form*. If the retained evidence also determines the public result, admissible schedules recover the same shared account. The proposal identifies objective reality with this invariant organization: the public answer does not depend on who repaired first.

Causal order and repair answer different questions. The first records which encounters depend on

which others; the second explains how different routes can settle on a common account of them.

4. From a shared world to a world for someone

Give each observer local state, ports through which it exchanges information, records it can reread, and permitted repair moves. Observers share some data without sharing everything. One may encode a length in centimetres and another in millimetres. They agree when their interpretations match under the legitimate translation.

Agreement preserves meaning across an interface without requiring identical storage or perspectives. The public world is the organization these partial accounts can sustain together.

Consider how that idea applies to seeing the cup. Different processes respond to contours, colour, position, and the possibilities of a reach. No complete cup travels through the eyes. The hidden side supplies no current image of itself; its persistence belongs to the model's expectations.

For these processes to concern one cup, the position used for reaching must fit the position assigned to its outline. Colour need not become a copy of shape. Their different contributions must support one usable world.

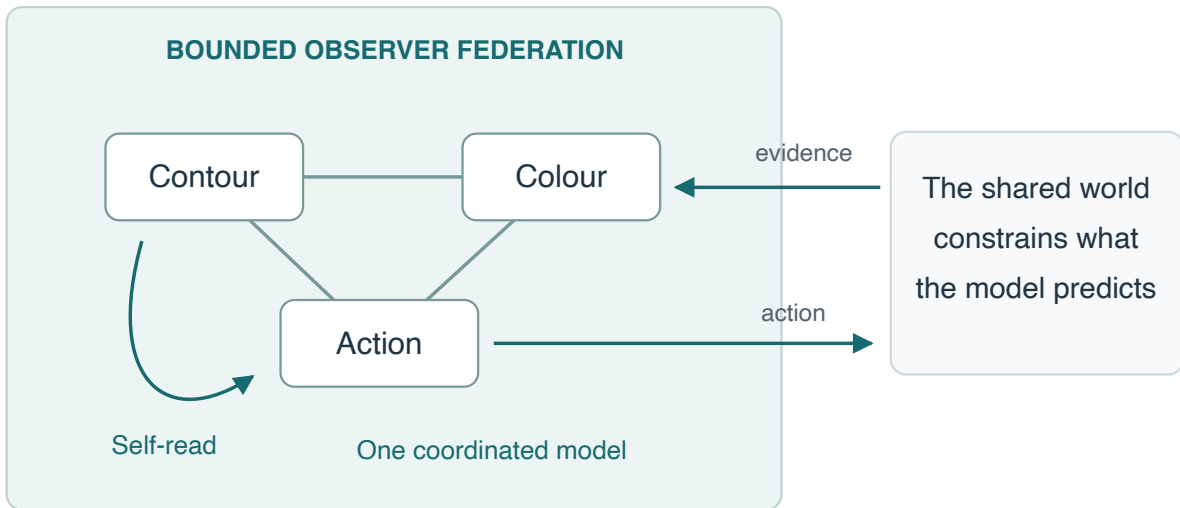


Figure 2: Partial contents constrain one another within an observer federation. Their coordinated self-reading supplies the proposed lived model; shared evidence constrains what that model can sustain.

A brain can maintain this organization while its components continue changing. The relevant stable result is a coordinated interpretation that can be committed, reread, and used.

The phenomenal proposal is that this enacted world model is the seeing. “Enacted” means that its content governs the organism’s own ongoing activity; the word is borrowed from the enactive tradition (Varela, Thompson, and Rosch 1991) and used in this narrower sense.

The cup’s presence includes what the observer can expect, remember, compare, and reconsider. There need be no completed inner picture followed by another act in which a spectator inside the head sees it.

The same organization can be active when the apparent cup is absent. A hallucination is real as an episode and mistaken as an account of the room. The model exists as an occurrence; its reference to a cup can fail. Construction therefore does not imply accuracy. The organism's continuing encounters constrain which interpretations survive.

We have reached the intended identification: the model's meaning in the observer's own activity is its lived content. Functional coordination alone does not prove this identity. To see what follows from it, we must distinguish the existence of experience from the distinctions an experience must contain.

5. Experience as basic

Start with the occurrence you are trying to explain. There is something it is like for you to see the cup. Calling that occurrence subjective identifies whose encounter it is. Calling its redness qualitative identifies a feature of how that encounter is.

Here *qualitative character* means what an episode is like for its subject. It need not involve words, a judgement about experience, or human sensory richness. Within this meaning, an actual subjective experience with no qualitative character would be an experience with nothing it is like to have it. That contradicts the starting description.

This establishes a conceptual consequence of accepting an experience. It does not show that every information-processing system has one. The phrase "the owner of an experience" already presupposes an experience. The owner is the unit whose activity constitutes that occurrence; no second observer needs to inspect it to make it present.

Two episodes can have exactly the same qualitative character. Counting their occurrences would not establish different contents.

Taking experience as more fundamental than spacetime changes where this explanation begins. The theory starts with experiencing activity and reconstructs spatial and temporal relations from the underlying organization.

This differs from Russellian monism (Russell 1927; Chalmers 2015), on which experiential properties are the categorical bases of dispositions that physics describes only structurally. There physical spacetime is presupposed and the intrinsic nature of what occupies it is in question. Here the experiential activity is the self-reading of bounded observers, and spacetime is reconstructed from their shared relations.

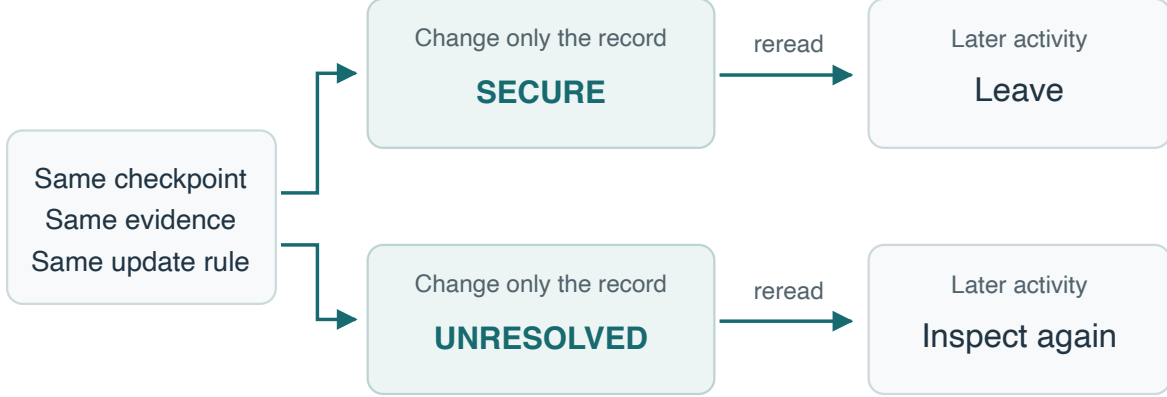
Explanatory priority alone leaves two questions unanswered: which organized activities constitute subjects, and why should their qualitative contents differ when their own responses differ? Saying that experiences are basic does not make every collection of underlying events one subject. Nor does it establish that every causal distinction is felt by that subject.

The observer proposal answers through two further commitments. A unity rule selects the experiencing unit. An identity relates that unit's qualitative content to the meaning enacted in its committed self-reading. These are the premises connecting an operational observer to an actual subject.

The nontrivial necessity argument concerns the second commitment. A subject whose own activity depends on distinctions cannot preserve that activity with one interchangeable lived content, if that content is what carries those distinctions. The following two steps make this statement precise.

6. Operative distinctions

Consider a self-reading system that retains either “secure” or “unresolved” after inspecting a door. Its later inspection policy depends on the record. Substitute one reachable record for the other after writing and before rereading, while holding the rest of the checkpoint fixed.



A downstream causal effect counts. A copied label does not.

Figure 3: Change only the committed record, then compare the same system’s later activity. Leaving and inspecting again are different continuations. An unused copy or an external audit label would not establish the required self-reading.

Let R be a finite set of admissible committed records. Let c specify the remaining checkpoint state and continuation conditions, and let $F_c(r)$ be the subsequent response after rereading record r . A response includes endogenous activity, rather than just a copy of the intervened record.¹

Group records that are interchangeable in every admitted context:

$$r \sim r' \iff F_c(r) = F_c(r') \text{ for every admitted } c.$$

Write $[r]$ for the group containing r , and $Q = R/\sim$ for the set of groups. Different spellings of “secure” might share a group. “Secure” and “unresolved” cannot share one if they change the inspection policy.

First result. Nontrivial self-reading requires $|Q| \geq 2$. If there were only one group, every record would have the same response in every admitted context. Rereading could make none of the stipulated differences.

Suppose all relevant record influence passes through an internal state $E(r)$, so that

$$F_c(r) = G_c(E(r)).$$

Equal internal states supplied to the same rule give equal responses. Therefore different responses require different internal states. The record’s effective distinction must survive in the system that

¹For stochastic dynamics, take the response to be its probability distribution.

uses it. A faithful compressed model must preserve it too.

This is the precise operational sense in which the observer must model a difference. It does not require an inner picture. It requires distinct operative states. An unmodelled channel bypassing E would invalidate the assumed factorization; causal relevance alone does not prove that an arbitrarily chosen representation carries the influence.

7. The necessity of lived distinctions

Apply the argument to an experiencing unit. The premises are specific:

1. **An actual subject.** A selected unit U has admissible subjective episodes corresponding to the records in R . Write $\varphi(r)$ for the qualitative content of the episode corresponding to r . This is the experiential and unity premise.
2. **Self-reading matters.** At least two records yield different continuations in some admitted context, as in the door example.
3. **Lived content carries the relevant meaning.** With the remaining conditions fixed, the record affects the declared continuation through the unit's enacted qualitative content. This is the consequence of the proposed phenomenal identity used by the proof.

The third premise says that two episodes with exactly the same relevant lived content cannot have different record-dependent continuations when the rest of the context is fixed. Equivalently, there are continuation rules H_c such that

$$F_c(r) = H_c(\varphi(r)).$$

This concerns the declared self-reading channel. It does not assert that every process in an organism is consciously experienced. Under the proposed identity, content consists of the enacted response class and its relations, so the factorization follows. The factorization is not a consequence of experience preceding spacetime alone.

Before the symbols, consider the contradiction. Suppose “secure” and “unresolved” make the unit continue differently, but its complete relevant lived content is identical in both cases. The same content would then have to produce different responses under the same continuation conditions. That contradicts the third premise.

Conditional necessity theorem. Let $\Phi = \varphi(R)$ be the repertoire of qualitative contents for these episodes. Under the three premises,

$$|\Phi| \geq |Q| \geq 2.$$

Proof. If $\varphi(r) = \varphi(r')$, the factorization gives $F_c(r) = F_c(r')$ for every admitted c . Therefore $[r] = [r']$. One qualitative content can correspond to only one response class. Every response class has at least one record and hence at least one corresponding qualitative content. There must be at least as many qualitative contents as response classes. Nontrivial self-reading supplies at least two classes. This proves the inequality.

The conclusion concerns possible contents across the admitted episodes. It does not require two qualities to be felt simultaneously. An actual episode has a qualitative character under the first premise; its possible alternatives must preserve the distinctions proved here. One generic, interchangeable “presence” cannot do the work of a differentiated lived world.

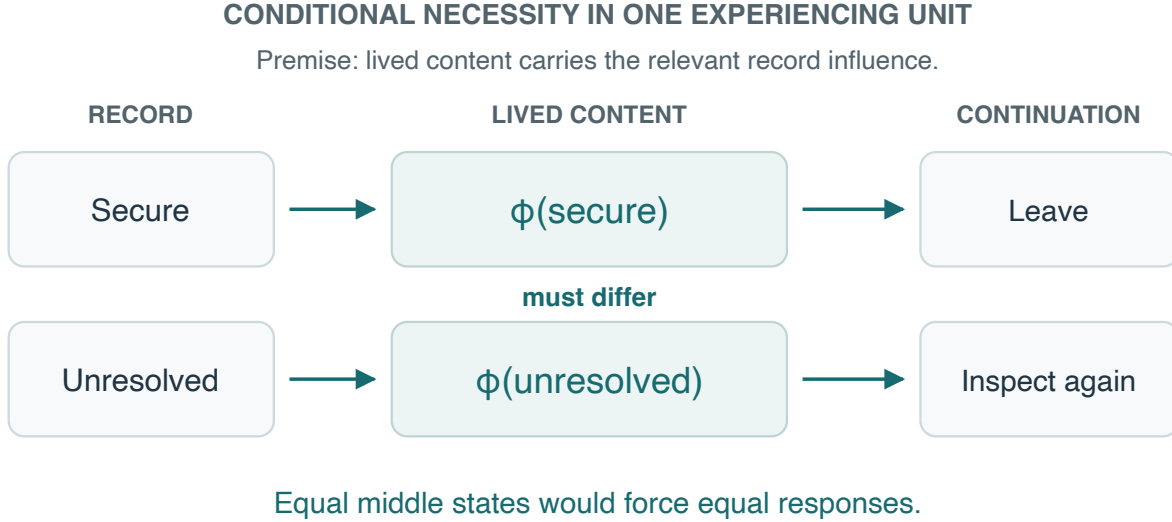


Figure 4: Conditional on the selected subject and the identity linking lived content to record-dependent activity, different continuations require different qualitative contents. The experience premise identifies the middle states as lived; the proof shows why they cannot be merged.

Each premise has a separate role. Without the first, the calculation concerns internal states without establishing feeling. Without the second, it forces no pair of different contents. Without the third, a constant experience could accompany different responses carried by some other state. Experiential primacy alone does not exclude that last arrangement.

The result explains the necessity claimed here. Experience is accepted as basic; its organization is constrained. Within the identity, removing the lived distinctions while preserving the observer's complete relevant activity is impossible. That is a consequence of the premises, rather than a derivation of experience from equations that never mention it.

8. Description and constitution

A novel can describe a frightened person. Reading it may produce an actual experience of imagining that person. The character does not thereby acquire a separate experiencing unit. The representation is part of the reader's activity; an additional subject would require an organization that qualifies in its own right.

The proposed brain model must therefore do more than portray a conscious person. Its retained states must enter the organism's own continuing self-reading. Under the phenomenal identity, that realized organization constitutes the experience. It need not produce another experience behind the representation.

The same activity can thus describe one thing and constitute another. The brain represents a person encountering a cup; the organized qualitative composition is, on this account, the encounter itself. Requiring another act to make that composition present would repeat the original demand inside its proposed answer. The identity rejects that extra step explicitly, rather than deriving its rejection from computational fidelity alone.

This supplies an eligibility condition for "description is existence": a description must realize the

organization constitutive of the particular thing claimed to exist. Words describing a memory changing a decision do not supply that dependency. A retained record whose alternatives change the decision can supply it. Chalmers's (2011) account of computational implementation makes a related distinction between formal description and causal structure.

Let s be an underlying state and $C(s)$ its description at the level of the brain's model. For each admitted operation a , let f_a be the underlying continuation and G_a the corresponding model update. A faithful realization requires a fixed mapping satisfying

$$C(f_a(s)) = G_a(C(s))$$

throughout the admitted states and operations. The mapping must cover the model states being claimed and identify their records and ports. The operations include controlled changes to those records. The correspondence is specified before testing; it cannot be fitted separately to each observed history.

Repeated substitution proves that the same correspondence holds for every finite admitted sequence. Thus the model's intervention-dependent responses are realized in the underlying activity. A matching movie of one sequence cannot establish that conclusion. Approximate realizations require explicit tolerances; the exact equation does not guarantee their accuracy.

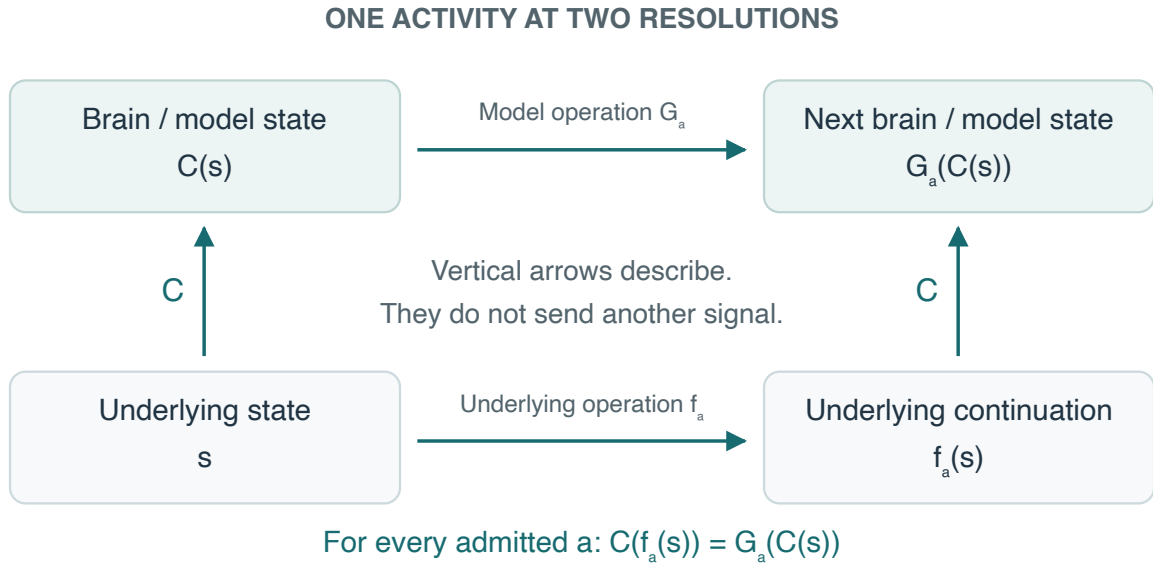


Figure 5: A fixed description map relates underlying activity to the brain's model. The equation must hold for each admitted operation, including the relevant record interventions. The vertical arrows describe states; they are not additional causal signals.

This proves preservation of the specified organization, given its causal implementation. Transferring the phenomenal claim also requires preservation of the experiencing unit and the relevant self-reading relations. Matching outward behaviour alone does not show that. The phenomenal identity says which realized organization constitutes experience; the mapping tests whether that organization is present.

A simulated fire can be a real computational process without burning the host machine. The

model represents a cup and, under the identity, instantiates an experience. The existence of that experience does not establish a cup on the table.

The brain’s construction of a percept is an organization of the underlying activity throughout. It sends no second message to a lower room where the description becomes alive.

The clocks must remain distinct here too. Underlying dependency depth, elapsed brain time measured by a physical clock, and experienced duration are different quantities. Felt duration belongs to the proposed lived content, including its organization of memory and expectation. Neither counting repairs nor counting neural oscillations alone derives it. Explanatory priority therefore adds no earlier instant at which qualia wait for neurons to act.

The further hypothesis of *intrinsic realization* identifies a complete inhabited causal structure with actuality. Its encounters need no external clock to play them. Combined with the phenomenal identity, the qualifying encounter is already experience. Mathematical consistency alone establishes neither commitment.

The basis of qualitative character is therefore located at the underlying observer level. A particular human percept depends on the organization of its selected whole, without requiring a complete human redness in each component. Spacetime is the further public reconstruction described in section 3.

9. The character of qualities

The necessity theorem establishes a repertoire of distinctions. It does not derive the particular character of redness or pain. To address that question, the account must explain relations among contents.

Using a separating measure D of response difference, define

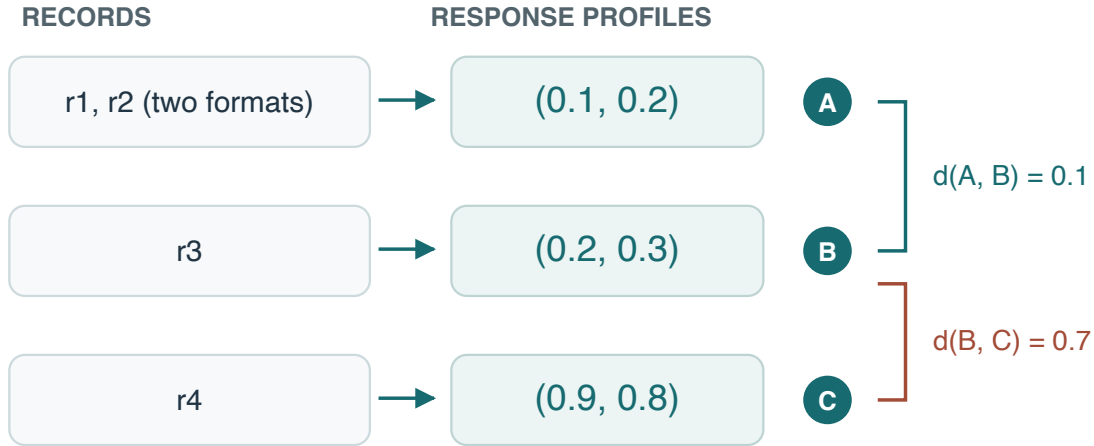
$$d(r, r') = \sup_c D(F_c(r), F_c(r')).$$

This measures the largest difference any admitted context reveals. Zero distance means that the records are response-equivalent. On the set of response classes it gives a metric, possibly allowing infinite distances.

For example, let three records yield inspection probabilities $(0.1, 0.2)$, $(0.2, 0.3)$, and $(0.9, 0.8)$ in two contexts. Using the largest coordinate difference gives distances 0.1, 0.8, and 0.7. The first two records can belong to distinct classes while being close in response.

Under the stronger identity, a quality’s character consists in its enacted class and its structured relations within the model. Colour involves related possibilities of discrimination and expectation. Pain involves further relations of protection and aversion. An isolated state label specifies little of this organization.

A wavelength does not arrive with “red” written on it. The substrate supplies permitted states and transitions, which we can call its syntax. The observer’s organization gives those possibilities semantic content through their role in its own encounter. Shared constraints prevent that contribution from becoming arbitrary invention.



Equivalent records share a cell; different cells can still be close.

Figure 6: Records acquire a geometry through their continuation responses. Equivalent encodings form one class; distinct classes can remain close. The example illustrates response geometry rather than a derived human colour space.

A surface can afford a foothold to one creature and present a wall to another. They can agree about the geometry while having different real relations to it. Meaning belongs to those relations as well as to the shared measurements.

The response contexts and metric must be fixed independently of the resemblance we hope to find. The proposed connection gains evidence only if the resulting geometry tracks independently assessed experiential distinctions. Choosing the test to reproduce a desired colour space would explain nothing.

10. The experiencing unit

The proof concerns a selected experiencing unit. Agreement about a cup does not by itself establish such a unit: several computers can exchange compatible descriptions without that fact deciding whether there is one experience or several.

Integrated information theory (Tononi 2008) proposes a test: partition the system and measure what the cut destroys. Suppose two bits jointly determine whether an action occurs through their parity. A part retaining only one bit cannot determine whether the bits differ when both may vary. Replacing the missing channel with a fixed value changes some responses.

Replaying the intact signals is insufficient as a control. It may reproduce one history while failing on alternative records. The comparison must preserve the declared conditions and test the same family of interventions.

These tests assess functional integration. The assignment of an experiencing unit remains a further postulate, here as in that theory, where its exclusion postulate does that work. It must specify which candidate owns an episode and address overlapping candidates and ties.

This is the combination problem for panpsychism (Seager 1995; Goff 2017) in the present setting.

The brain's components do not automatically become a crowd of little subjects, and the shared world does not automatically become one cosmic subject. The observer ontology can allow minimal experience widely while requiring particular organization for a unified human encounter. Subject boundaries are a substantive part of that account.

11. Public records and reconstruction

Two different questions are often combined under privacy. Can another observer learn my experience's content? Can learning it make that very occurrence theirs? The first question is about available information.

Let s be an admissible state, $P(s)$ the record available through a declared public interface, and $q(s)$ the owned content to be recovered. A decoder D would have to satisfy

$$D(P(s)) = q(s).$$

Reconstruction criterion. Such a decoder exists exactly when

$$P(s) = P(t) \implies q(s) = q(t).$$

Proof. If the same public record can accompany two different contents, one input would require two decoder outputs. That is impossible. Conversely, if every state with a given public record has the same content, assign that content to that record. This defines a decoder on all records that can occur.

Take $s = (p, h)$, with $P(p, h) = p$ and $q(p, h) = h$. The states $(0, 0)$ and $(0, 1)$ have the same public bit and different owned contents. No computation using only that bit can distinguish them.

This repeats the logical principle behind the necessity theorem: the input to a faithful map must preserve the distinctions its output needs.

The privacy conclusion is conditional on omission. Publishing both bits makes the example decodable. Later behaviour may also reveal a hidden state. A finite interface can transmit a finite state completely; boundedness alone does not prove secrecy.

An impoverished public description can therefore fail to determine an experience. A complete description need not fail. The second question survives even perfect information: does knowing the content make its original occurrence yours?

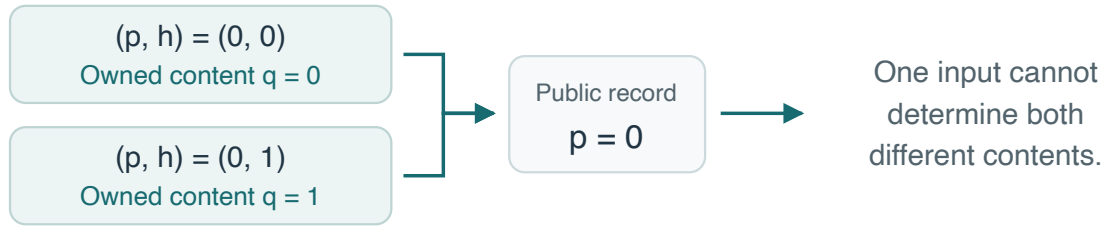
12. Knowing and undergoing

A pain can be known to everyone in the room and remain the patient's pain. What makes it the patient's occurrence is its place in that patient's continuing activity.

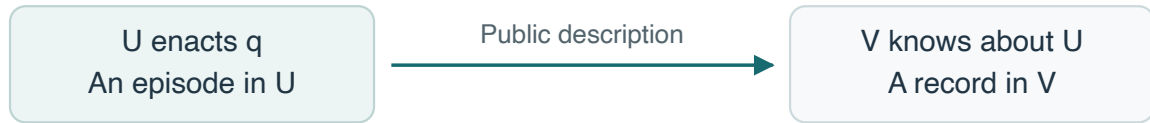
Distinguish a qualitative pattern from an episode instantiating it. Two observers may instantiate the same pattern. Their episodes can nevertheless have different memories, boundaries, and consequences. "The same red" need not mean one numerically identical event.

Suppose the unit rule assigns an episode e a unique owner U . Another distinct unit V receiving a description does not, merely by learning it, become U or change that assignment.

A RECORD MAY OMIT A DISTINCTION



KNOWING DOES NOT TRANSFER THE OCCURRENCE



If V enacts a matching quality, that is an episode in V.

Figure 7: A public interface can erase a distinction between owned contents. No decoder can recover both from the identical record. A richer interface may reveal the content without transferring the original episode.

If V enacts a matching content, there is an episode in V. If a connection instead creates one experiencing whole, the boundary selecting the subject has changed. Neither operation gives the same unchanged episode two distinct owners under the stipulated unique-ownership rule.

This is privacy of occurrence. It follows from the enactment identity and the ownership rule; the rule itself needs justification. It permits matching qualities, communication about experiences, and the possibility of a collective subject.

Jackson's (1982) Mary makes the distinction concrete. Give her a complete atlas of colour experience, specifying every candidate state and relation, including what a red episode would do in her own continuing system. The atlas can be complete while she has never occupied that state.

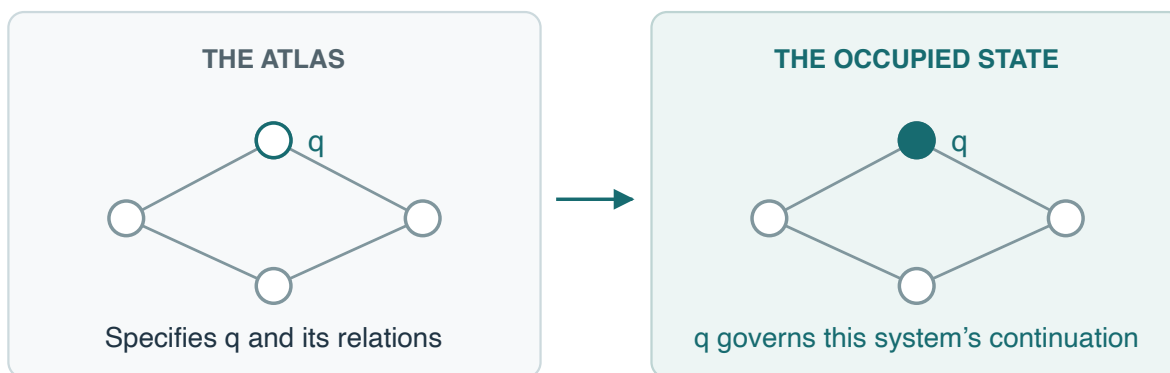
When she sees red, the described organization becomes her own activity. Her "So that is what red is like" connects a known description with an enacted content. Under the identity, the gain is acquaintance through realization.

Perry's (1979) account of self-locating knowledge offers a related point: a complete map does not by itself establish "I am here" for its reader. A representation can specify a relation without putting its reader into it. Identifying this with phenomenal acquaintance is a further philosophical step.

To model the observer faithfully, preserve its operative distinctions. To undergo the corresponding episode, enact the appropriate organization. Fidelity of description and ownership of an occurrence are different requirements.

13. Scope of the argument

The hard problem asks why physical activity should be accompanied by experience. The proposed answer begins with subjective activity and identifies its qualitative content with the meaning en-



Knowing the atlas does not enact every state it describes.

Figure 8: An atlas can specify a state completely. Occupying that state places it in the observer's own continuation. Receiving the description and undergoing the described episode are different relations to the content.

acted there. Spacetime is recovered from the underlying organization where the reconstruction conditions hold.

Under the experiential mediation premise, lived content must preserve the observer's response distinctions. Under the ownership rule, a public account can preserve those distinctions without transferring the episode. Causal realization must preserve the relevant organization for either consequence to apply.

Accepting actual experience establishes qualitative character in the stated sense. The response theorem establishes necessary operative distinctions. The identification and unity premises connect those distinctions to a subject's experience. Causal realization says when the described organization is instantiated. Explanatory priority to spacetime provides the proposed grounding; it supplies neither connecting premise by itself.

Chalmers's (1995) objection remains available to someone who rejects the identity: why could the complete functional organization not exist without feeling? The operational equations alone leave that interpretation unconstrained. The present account gives a definite answer by identifying subjective presence with enacted content. Its success depends on whether that identification explains experience and survives evidence, rather than on a definition compelling assent.

14. Conclusion: the answer belongs to the reading

Look at the cup again.

Its colour is available with its shape and its possibilities. What you see changes what you expect; uncertainty can make you look again. Under the proposed identity, these are relations within your lived encounter. Remove the distinctions they require and the same self-reading activity cannot continue unchanged.

Your experience does not become qualitative only after another observer notices it. Being this actual encounter supplies its character for you. The theorem explains why that character must be differentiated across the possibilities your activity can use.

Describe the encounter perfectly to someone else. You may convey every distinction and even the means to realize a matching episode. Your occurrence nevertheless belongs to the activity constituting your continuing perspective. Its privacy is compatible with being understood.

At the neural scale, this is an integrated percept. At the proposed underlying scale, it is the organized self-reading through which the percept exists. The account gives experience and public reality a common basis while preserving their different scopes.

The universe creates meaning through these encounters. What observers can sustain together forms a shared world; what a selected experiencing unit enacts is its lived world. You participate in both. The meanings you make are constrained by reality because your activity belongs to that reality.

The same relation holds for these sentences. Their meaning becomes your understanding through the reading occurring in you. An account of understanding becomes part of an actual act of understanding.

Why do qualia exist? On this account, they are the character of a world being lived by its owner. Their differentiated organization is necessary wherever that lived content carries the observer's own distinctions. Describing it does not take the place of undergoing it.

You are where this instance occurs. The world becoming meaningful here is the experience.

References

- Bombelli, Luca, Joohan Lee, David Meyer, and Rafael D. Sorkin. 1987. "Space-Time as a Causal Set." *Physical Review Letters* 59: 521–524.
- Chalmers, David J. 1995. "Facing Up to the Problem of Consciousness." *Journal of Consciousness Studies* 2: 200–219.
- Chalmers, David J. 2011. "A Computational Foundation for the Study of Cognition." *Journal of Cognitive Science* 12: 325–359.
- Chalmers, David J. 2015. "Panpsychism and Panprotopsyism." In *Consciousness in the Physical World*, edited by T. Alter and Y. Nagasawa, 246–276. Oxford: Oxford University Press.
- Goff, Philip. 2017. *Consciousness and Fundamental Reality*. Oxford: Oxford University Press.
- Jackson, Frank. 1982. "Epiphenomenal Qualia." *Philosophical Quarterly* 32: 127–136.
- Lamport, Leslie. 1978. "Time, Clocks, and the Ordering of Events in a Distributed System." *Communications of the ACM* 21: 558–565.

- Nagel, Thomas. 1974. "What Is It Like to Be a Bat?" *Philosophical Review* 83: 435–450.
- Perry, John. 1979. "The Problem of the Essential Indexical." *Noûs* 13: 3–21.
- Russell, Bertrand. 1927. *The Analysis of Matter*. London: Kegan Paul.
- Seager, William. 1995. "Consciousness, Information and Panpsychism." *Journal of Consciousness Studies* 2: 272–288.
- Tononi, Giulio. 2008. "Consciousness as Integrated Information: A Provisional Manifesto." *Biological Bulletin* 215: 216–242.
- Varela, Francisco J., Evan Thompson, and Eleanor Rosch. 1991. *The Embodied Mind*. Cambridge, MA: MIT Press.